Sheet 53 Mon. 22 March 2027 Complete a two-way table

1. Warm-up · Flash Calculate 35% of 80.

2. Core · Method 200 customers: 60% women; 25% of the women and 50% of the men buy online. Complete the two-way table.

3. Core + · Check A table states "row total: 45 $+ 55 = 100$" but "column total: $60 + 30 = 90$". What can be concluded?

4. Back · True/false True or false? Justify. After three "heads" in a row, the probability of getting "tails" on the next toss is greater than $\frac{1}{2}$.

5. Running thread · Flash ABC right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.

Sheet 54 Tue. 23 March 2027 Real data: processing a file and filtering (AND, OR, NOT)

1. Warm-up · Flash In a file of 1,000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy A & B. How many satisfy " A AND B" ? "A OR B" ?

2. Core · Method With the same data: how many satisfy "NOT A" ? "A AND NOT B" ?

3. Core + · Reading code What is the final value of compte ?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```

4. Back · Flash Give the decimal expansion of $\frac{7}{8}$ then the irreducible fraction equal to 0.12.

5. Running thread · Formula → table $f(x) = x^2 - 2x$. Complete a table for $x \in \{0\,;\,1\,;\,2\,;\,3\}$.

Sheet 55 Thu. 25 March 2027 Critical reading of media formulations

1. Warm-up · Flash A 10 % increase on 40: what final value?

2. Core · Critical thinking An article states: "60 % of accidents involve sober drivers". Can we conclude that driving sober is dangerous?

3. Core + · True/false True or false? Justify. "Two out of three smokers are men" means that two out of three men smoke.

4. Back · True/false True or false? Justify. The median of a series is always equal to its mean.

5. Running thread · Reading code What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

Coordinate geometry: vectors and lines review

Sheet 56 Tue. 30 March 2027 Vectors and collinearity: review

1. Warm-up · Flash $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ & $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.

2. Core · Method Are points $A(-1;2)$, $B(1;3)$ and $C(5;5)$ collinear?

3. Core + · Method $A(-1;2)$, $B(3;1)$, $C(4;4)$. Determine D such that $ABCD$ is a parallelogram.

4. Back · Flash In a class of 24 pupils, 18 are half-boarders. What proportion is this as a percentage?

5. Running thread · Prove Show that $(n+1)^2 - (n-1)^2 = 4n$ for every integer n. What can we conclude about the difference of the squares of two consecutive odd numbers (n even)?